

GasGo

Gas Delivery Platform

System Design Requirements

Version 1.0 · Confidential

On-demand LPG cylinder delivery & exchange web application

1. Product Overview

GasGo is an on-demand LPG (liquefied petroleum gas) delivery and exchange web application. Users request a gas cylinder fill or exchange from their location; the platform matches them with the nearest available station, assigns a rider, and provides real-time delivery tracking. Station owners set their own prices per cylinder size. The platform earns a commission on each completed order.

Core value propositions

- **Convenience** — Users never leave home — gas is brought to their door within a targeted SLA.
- **Price transparency** — Station prices are shown upfront; users can compare before ordering.
- **Gas exchange** — Old cylinders are exchanged for new filled ones at a lower price, eliminating the need to source a replacement.
- **Real-time tracking** — Live rider location and status updates at every stage of the order.
- **Station empowerment** — Stations control inventory, pricing, and see analytics without manual reconciliation.

2. System Actors

Actor	Interface	Primary responsibility
User	Mobile-first web app	Browse stations, place orders, track delivery, rate rider
Rider	Mobile web / PWA	Accept orders, collect & deliver cylinders, confirm via OTP
Station	Web dashboard	Set prices, manage inventory, view order queue, analytics
Admin	Web dashboard	Approve stations/riders, platform pricing, global metrics

3. Platform Architecture

The diagram below maps the four actor layers and how they connect to the core backend services.

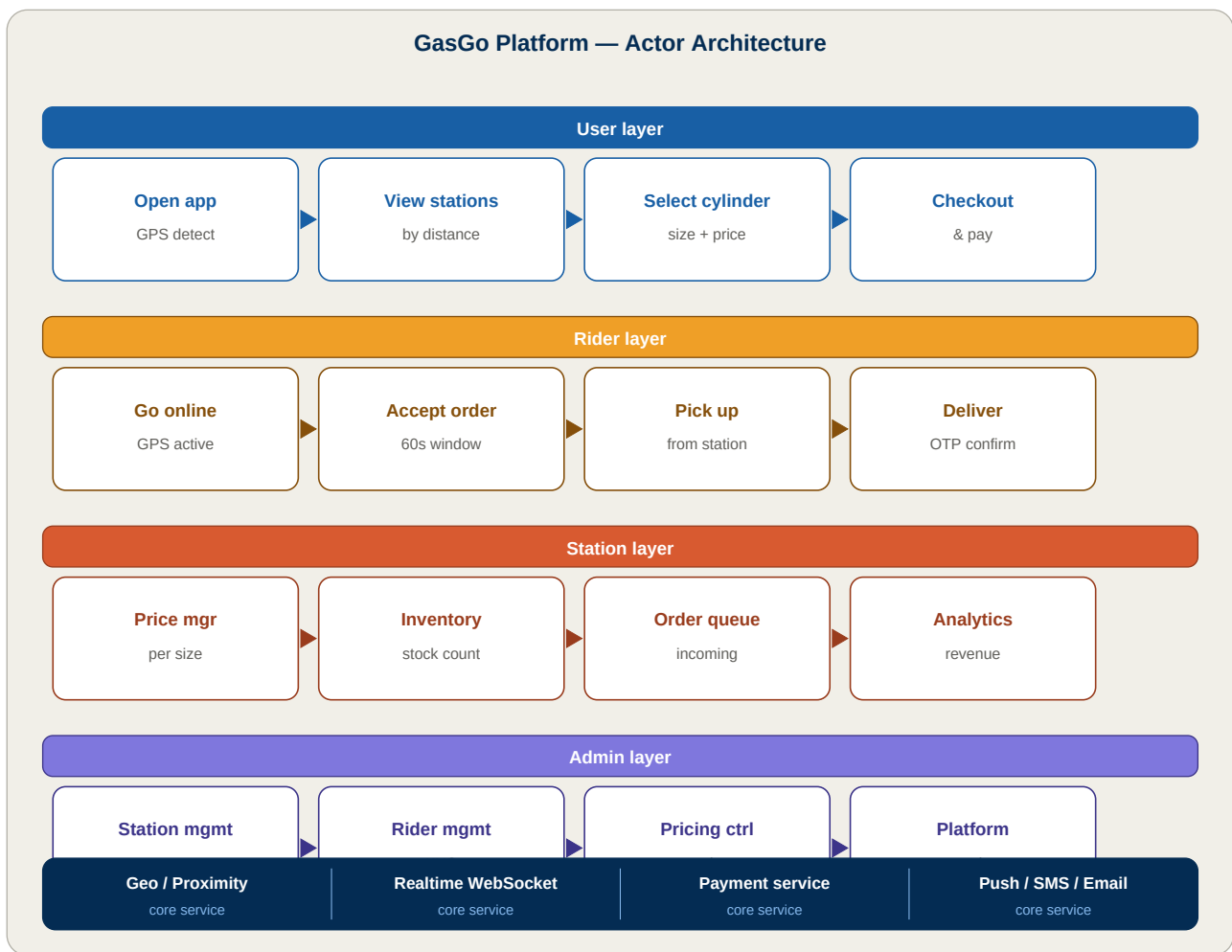


Figure 1 — GasGo platform actor architecture and core services

4. User App — Functional Requirements

4.1 Onboarding & Authentication

- Register with phone number + OTP verification.
- Profile stores: name, saved delivery addresses, payment methods, cylinder order history.
- Location permission requested on first launch; fallback to manual address entry.
- Social login (Google) as optional fast path in v2.

4.2 Station Discovery

- On app open, GPS coordinates are detected and sent to the Geo service.
- Stations are returned sorted by straight-line distance (closest first).
- Results filtered to active stations with stock in at least one cylinder size.
- Each station card shows: name, distance, rating, available sizes, and starting price.
- Users can manually browse other stations outside the default radius.

4.3 Cylinder Selection

- Standard sizes: 3 kg (small home), 6 kg (standard home), 12 kg (large/commercial).
- Each size is shown as a selectable card with the station's current price.

- Tapping a size opens a detail sheet: brand, fill type, weight, and availability.
- Users can enquire about stock before placing an order (no commitment).

4.4 Order Types

- Fill order — user already has the cylinder; rider collects, fills at station, returns.
- Delivery order — station sends a new pre-filled cylinder directly to the user.
- Exchange order — rider swaps the user's old cylinder for a new filled one.
- Order type is selected at checkout. Price displayed updates per selection.

4.5 Gas Exchange Flow

- User selects "Exchange" and declares the size of their old cylinder.
- Exchange price (set by the station, always $\leq$ fill price) shown upfront.
- Rider picks up old cylinder from user's doorstep, delivers it to the station.
- Rider collects new pre-filled cylinder and returns to the user.
- Station logs the returned cylinder into the "needs refill" queue.

4.6 Checkout & Payment

- User confirms: delivery address, order type, cylinder selection.
- Total = cylinder price + platform delivery fee.
- Payment methods: mobile money (MTN/Vodafone Cash), debit/credit card, cash on delivery.
- On payment confirmation, order is created with status = Pending.
- Payment held in escrow; released to station on delivery confirmation.

4.7 Live Tracking

- Once a rider accepts, user sees a live map with rider GPS position.
- Order status label updates in real time: Accepted → At station → En route → Delivered.
- ETA countdown displayed and recalculated every 30 seconds.
- Push notifications sent at every status transition.

4.8 Rating & Feedback

- After delivery, user is prompted to rate the rider (1–5 stars).
- Optional text comment for detailed feedback.
- Ratings feed into the rider's rolling average, visible on their profile.

5. Rider App — Functional Requirements

5.1 Onboarding & KYC

- Rider registers with: name, phone, national ID, vehicle type (motorbike / tricycle / van).
- Admin reviews and approves the KYC submission before the rider can go online.
- Background check confirmation stored against rider profile.

5.2 Online / Offline Toggle

- Rider switches status to Available; platform begins tracking GPS every 10–15 seconds.
- Going offline removes rider from the dispatch pool immediately.
- Riders can set a break duration to pause without fully going offline.

5.3 Order Assignment

- When an order is placed, the system finds the nearest available rider within a configurable radius (default 5 km).
- Rider receives a push notification with order summary and has 60 seconds to accept.
- If declined or timed out, the next closest rider is tried. After 3 failures, admin is alerted.

5.4 Standard Delivery Workflow

- Step 1: Navigate to the station address (in-app map with turn-by-turn navigation).
- Step 2: Confirm cylinder pick-up at station (station marks "ready for pickup").
- Step 3: Navigate to user's delivery address.
- Step 4: User enters 4-digit OTP on rider's screen to confirm receipt.
- Step 5: Rider optionally uploads a delivery photo.

5.5 Exchange Workflow

- Step 1: Navigate to user address — collect old cylinder.
- Step 2: Navigate to station — drop off old cylinder, collect new filled one.
- Step 3: Return to user address — deliver new cylinder.
- Step 4: OTP confirmation at each handoff point.
- The rider app shows a clear multi-step progress indicator for exchange orders.

5.6 Rider Dashboard

- Today's earnings (real-time), total trips completed, average rating.
- Active order map view with navigation shortcut.
- Full delivery history with order details and earnings breakdown.
- Payout history and bank/mobile money details management.

6. Station Dashboard — Functional Requirements

6.1 Price Manager

- Station sets a fill price and an exchange price per cylinder size (3 kg, 6 kg, 12 kg).
- Constraint enforced: `exchange_price <= fill_price` for the same size.
- Prices update in real time — next user session sees the new price.
- Full price change history is logged for dispute resolution.

6.2 Inventory Manager

- Station logs stock count per cylinder size.
- When a filled cylinder is dispatched, stock decrements automatically.
- Returned exchange cylinders enter a "needs refill" queue, tracked separately.
- Configurable low-stock alert threshold per size; notification sent when breached.
- Station can pause orders for a size when stock is 0.

6.3 Order Queue

- Three-column kanban view: Incoming → Active → Completed.
- Each order card: cylinder size, order type, rider name, customer address, timestamp.
- Station marks an order "ready for pickup" once the cylinder is prepared.
- Sound/visual alert on new incoming orders.
- Completed column shows today's orders; historical view with date filters.

6.4 Exchange Handling

- Dedicated view for cylinders currently at the station awaiting refill.
- Each returned cylinder tracked with: original order ID, size, return timestamp.
- Station marks cylinders as "refilled" to move them back to available stock.

6.5 Analytics

- Daily and weekly charts for: orders fulfilled, revenue, exchange vs. fill split.
- Most popular cylinder size by order count and revenue.
- Average delivery time (order placed → delivered).
- Platform commission deducted shown clearly in earnings summary.

6.6 Profile & Settings

- Station name, address, geolocation (set on registration, editable by admin).
- Operating hours per day of week; orders outside hours are blocked.
- Bank / mobile money account for payouts.
- Notification preferences: push, email, or both.

7. Admin Dashboard — Functional Requirements

Station management

- Review and approve new station registration applications.
- Suspend or permanently ban non-compliant stations.
- Set platform commission percentage per station (default or custom rate).
- Override station geolocation if incorrect.

Rider management

- Review KYC documents and approve or reject rider applications.
- View rider performance: trips, ratings, complaints, earnings.
- Suspend or ban riders with policy violations.
- Assign riders to specific zones or station clusters.

Pricing controls

- Set platform-wide delivery fee structure (flat or distance-based).
- Enable surge pricing multipliers during peak demand periods.
- Set minimum and maximum price caps per cylinder size to protect users.
- Emergency price freeze during supply shocks.

Platform metrics

- Total orders, GMV (gross merchandise value), average order value.
- Active stations, active riders, new user signups.
- Retention rate, complaint resolution rate, SLA breach percentage.
- Heatmap of order density by geography and time.

Dispute & support

- View flagged orders and customer complaints.
- Manually trigger refunds or order cancellations.
- Communicate with stations and riders via in-app messaging.

8. Order Lifecycle

Every order moves through a defined state machine. Status transitions trigger notifications to all relevant actors and update the realtime tracking view.

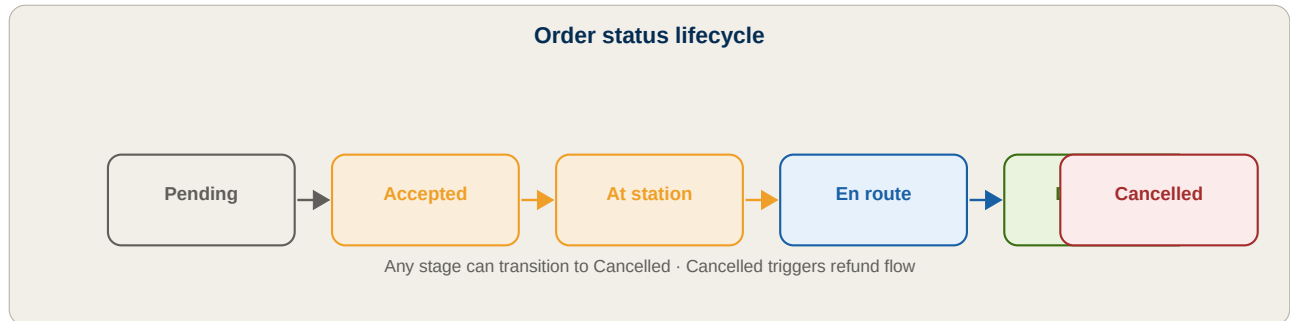


Figure 2 — Order status lifecycle

Status	Triggered by	Notifies
Pending	User completes payment	Station, nearest riders
Accepted	Rider accepts dispatch	User, Station
At station	Rider arrives at station	User, Station
En route	Rider departs station with cylinder	User
Delivered	User enters OTP	User, Station, Admin
Cancelled	User / Station / Admin action	All parties + refund trigger

9. Core Data Models

The three primary entities and their key fields. Foreign key relationships are shown between models.

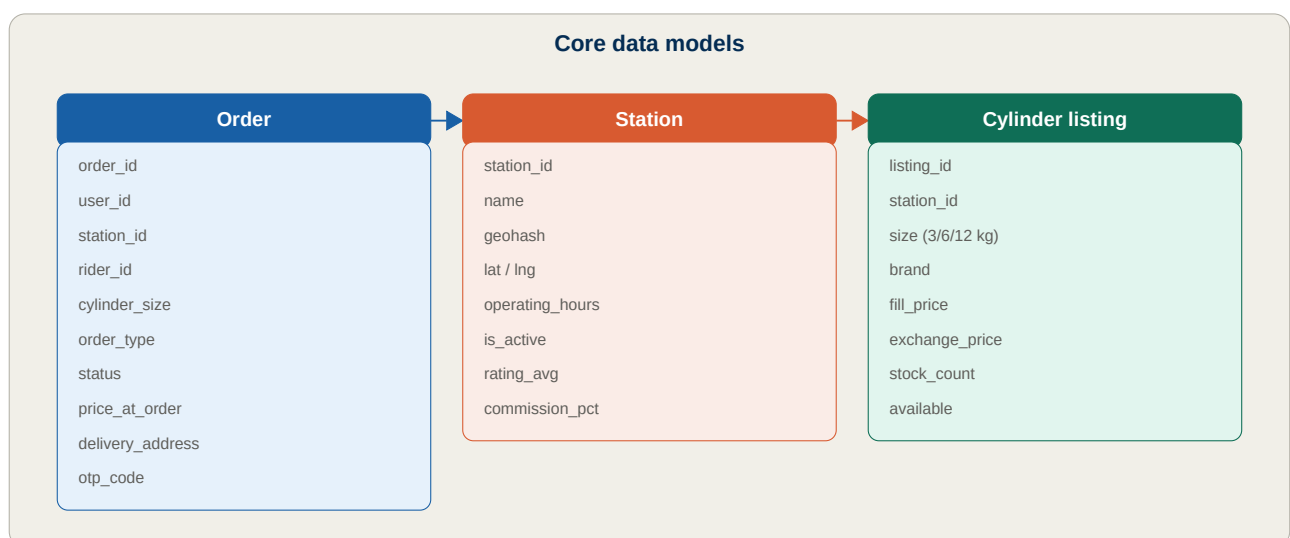


Figure 3 — Core data models: Order, Station, and Cylinder listing

Entity	Field	Type	Notes
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Order	order_type	ENUM	fill delivery exchange
Order	status	ENUM	pending accepted at_station en_route delivered cancelled
Order	price_at_order	DECIMAL	Snapshot of price at time of order — not live price
Station	geohash	VARCHAR(10)	Used for fast proximity queries (Geohash level 7)
Station	operating_hours	JSONB	{mon: {open: "08:00", close: "18:00"}, ...}
Cylinder listing	exchange_price	DECIMAL	Must be <= fill_price; enforced at application layer
Cylinder listing	stock_count	INT	Decrements on dispatch; increments when refilled and restocked

10. Core Backend Services

Geo service

- Accepts user GPS coordinates; returns stations within radius sorted by Haversine distance.
- Stations indexed by geohash (level 7 = ~150m precision) for fast bounding-box queries.
- Radius configurable per city; default 10 km with a 25 km hard ceiling.

Realtime engine

- WebSocket server (or Server-Sent Events fallback) for live rider location push.
- Rider GPS reported every 10–15 seconds while an order is active.
- Order status changes broadcast instantly to all subscribed clients.
- Rooms per order_id to scope broadcasts — no cross-order leakage.

Payment service

- Integrates with Paystack or Hubtel for Ghana mobile money and card processing.
- Payment held in escrow on order creation.
- Released to station account (minus commission) on status = Delivered.
- Refund triggered automatically on status = Cancelled, settled within 24 hours.

Dispatch engine

- On order creation, queries riders within radius ordered by distance.
- Sends push to nearest rider; starts a 60-second countdown.
- On timeout or decline, tries the next rider; logs each attempt.
- After 3 failed assignments, escalates to admin alert queue.

Notification service

- Sends push notifications (Firebase Cloud Messaging) for status transitions.
- SMS fallback (Arkesel or Twilio) for users without push enabled.
- Email for receipts, KYC outcomes, and payout confirmations.
- All notification templates managed centrally for easy copy updates.

OTP service

- Generates a cryptographically random 4-digit code per order on dispatch.
- Code expires after 30 minutes or once used, whichever comes first.
- Maximum 3 attempts before order is flagged for manual review.

11. Non-Functional Requirements

Requirement	Target
API response time (p95)	< 400 ms

Realtime location update latency	< 2 seconds end-to-end
Uptime SLA	99.5% monthly
Payment processing timeout	< 8 seconds
Order assignment time (rider found)	< 90 seconds
Mobile app first load (3G)	< 4 seconds
Data retention — order records	5 years (regulatory)
GDPR / data privacy	User data deletable on request within 30 days

12. Key Design Decisions

Station-set pricing

Prices differ between stations and are set by the station owner. The platform does not mandate a price floor (except via admin caps). This creates natural market competition between stations.

Exchange price constraint

The platform enforces at the application layer that a station's exchange price must always be $\leq$ its fill price for the same cylinder size. This protects users from being charged more for an exchange than a fresh purchase.

Closest station default

Station proximity is calculated from the user's live GPS on each session open, not from a saved address. Users can override to browse other stations.

Escrow payment model

Payment is captured on order creation and held in escrow. It is released to the station only after OTP-confirmed delivery. This protects users from non-delivery while ensuring stations are paid promptly.

v1 scope — delivery only

The "fill order" (collect from user, fill, return) is complex logistics. v1 ships with delivery and exchange only. Fill-and-return is scoped for v2 after operational learnings.

Geohash indexing

Stations are indexed by geohash at level 7 (~150 m cells). Proximity queries expand to neighbouring cells to avoid boundary edge cases, then sort results by true Haversine distance.

13. Phased Delivery Roadmap

Phase	Scope	Target
Phase 1 — MVP	User app (delivery + exchange orders), Station dashboard (price + inventory + order flow), Rider app (basic flow + order flow)	10-14 weeks
Phase 2 — Operations	Admin dashboard, analytics, surge pricing, fill-and-return order type, rider KYC verification, payout automation	6-8 weeks
Phase 3 — Growth	Referral programme, scheduled deliveries, loyalty points, multi-city geo expansion, station rating and review system	8-10 weeks